SUBJECT NAME: Cyber Security **Credit: 4**
SUBJECT CODE: BCAC703

COURSE OBJECTIVE:
The primary objective of this course is to provide students with a comprehensive understanding of the fundamental concepts, practices, and technologies in the field of Cyber Security. It aims to build a strong foundation in networking essentials, security principles, cryptography, and information security management, enabling learners to analyze and address real-world security challenges. Students will gain knowledge of the security triad (confidentiality, integrity, availability) and explore the phases of ethical hacking, risk assessment, and security auditing through case-based learning. The course emphasizes various cyber threats, such as malware, phishing, denial-of-service attacks, and social engineering, alongside hands-on exposure to countermeasures including firewalls, IDS/IPS, VPNs, and modern authentication mechanisms. Additionally, learners will be introduced to emerging technologies like AI-driven security, blockchain, and quantum cryptography. By the end of the course, students will be equipped with the skills to critically assess vulnerabilities, implement security policies, and contribute effectively to safeguarding digital assets in professional environments.

COURSE OUTCOME:

Course Outcome	
	DESCRIPTION
CO1	Understand the fundamentals of computer networks, OSI/TCP-IP framework, transmission media, and network devices, and analyze their role in secure communication with real-life case studies.
CO2	Explain the core concepts of Cyber Security, including the security triad (confidentiality, integrity, availability), hacking methodologies, and the role of hackers, supported with practical examples.
CO3	Apply cryptographic techniques and watermarking methods to secure digital data, differentiate between symmetric and asymmetric encryption, and evaluate the RSA algorithm for practical use.
CO4	Analyze and implement Information Security Management practices, including ISMS procedures, goals, and security audits, through case-based discussion.
CO5	Identify, classify, and evaluate various types of cyberattacks and technological defense mechanisms, and propose suitable security measures for real-world scenarios.

DETAILED SYLLABUS:

Module	Topics	Hours	Marks Distribution
Module 1: Networking Fundamentals for Cyber Security	Introduction to Networking, OSI & TCP/IP Framework, Transmission Media (with Case Study), Network Devices (with Case Study), TCP/IP Framework in detail, Subnet Masking (with Case Study), Three-way Handshaking, NAT, Different Protocols & their uses	14 Hours	20 Marks
Module 2: Cyber Security Foundations	Cyber Security Definition, Security Triad (CIA) with Case Study, Hacking & Ethical Hacking, Types of Hackers, Phases of Hacking (Reconnaissance, Scanning, Gaining Access, Maintaining Access, Covering Tracks), Cryptography & Watermarking and their use, Types of Cryptography, Symmetric vs Asymmetric Algorithms, RSA Algorithm explained	14 Hours	20 Marks
Module 3: Information Security Management	Introduction to ISM, Goals of Information Security, ISMS (Information Security Management System) working procedure, Security Policies, Risk Management, Information Security Audit (with Case Study)	10 Hours	15 Marks
Module 4: Cyber Attacks	Types of Attacks: Passive & Active Attacks, Malware, Phishing, Denial of Service, SQL Injection, Social Engineering, Insider Threats, Case Studies of Major Cyber Attacks	10 Hours	20 Marks
Module 5: Technological Aspects of Cyber Security	Firewalls, IDS & IPS, VPNs, Security Protocols (SSL, TLS, IPsec), Intrusion Detection Systems, Honeypots, Multi-Factor Authentication, Emerging Technologies in Cyber Security (AI, Blockchain in Security, Quantum Cryptography)	12 Hours	25 Marks
Total		**60 Hours**	**100 Marks**

BOOKS AND REFERENCES:

- Bhushan, M., & Rathore, R. S. (2017). Fundamentals of Cyber Security (Principles, Theory & Practices). *New Delhi: BPB Publications.*
- Shukla, S. K., & Agrawal, M. (2020). Cyber Security in India: Education, Research and Training. *Singapore: Springer.*
- Basta, A., Basta, N., Brown, M., & Kumar, R. (2018). Cyber Security and Cyber Laws. New Delhi: Cengage Learning India.
- Godbole, N. (2011). *Cyber Security*. New Delhi: Wiley India.
- Mukhopadhyay, D., & Forouzan, B. A. (2011). Cryptography and Network Security. New Delhi: Tata McGraw-Hill Education.